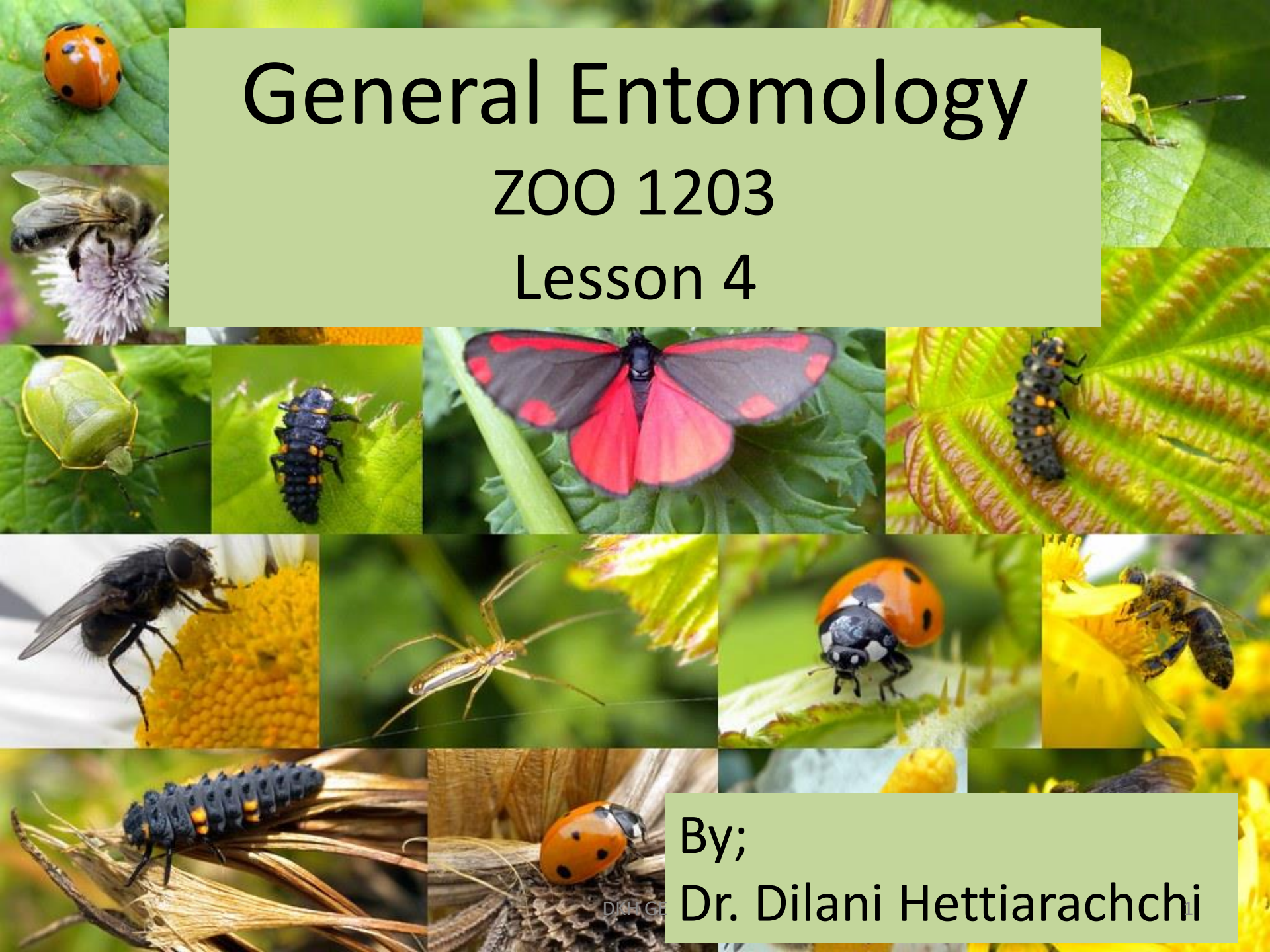


General Entomology

ZOO 1203

Lesson 4



By;

Dr. Dilani Hettiarachchi

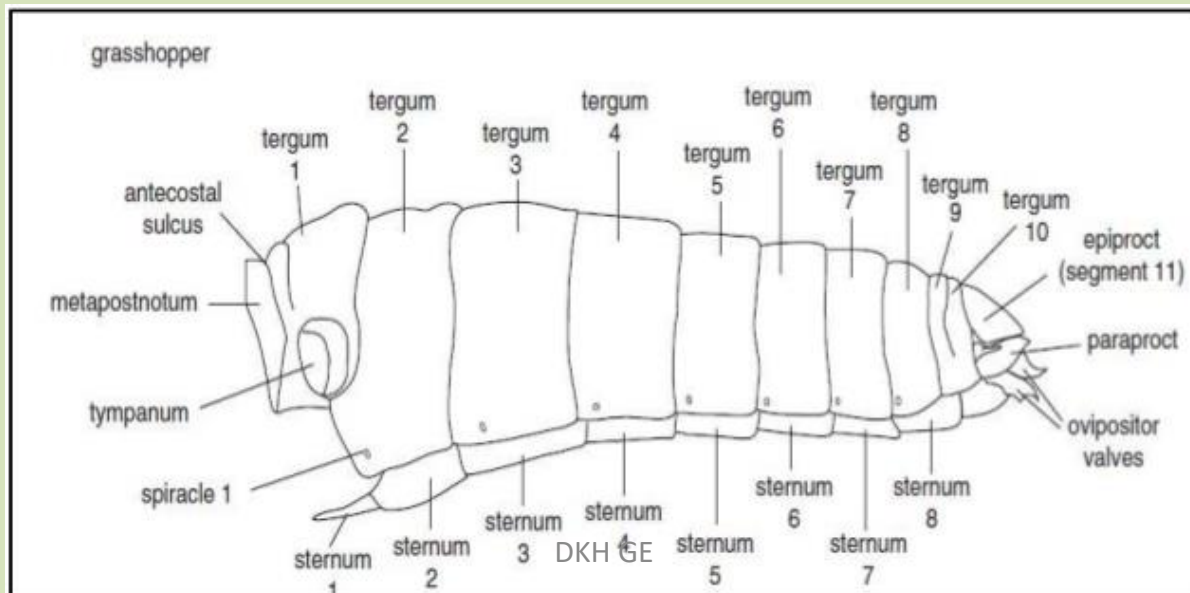
Lesson Objectives

After completion you should be able to;

- Identify the body regions of an insect
- Explain the various types of head appendages
- Identify the antennal types using the shape
- Describe different mouthparts of insects
- Discuss the adaptations of insect mouthparts to different mode of feeding

Segmentation

- Body is divided into a series of segments
- Somites or metameres
- Segments are heteronomous – different regions have different size, shape and other details
- Segments are grouped – tagmata

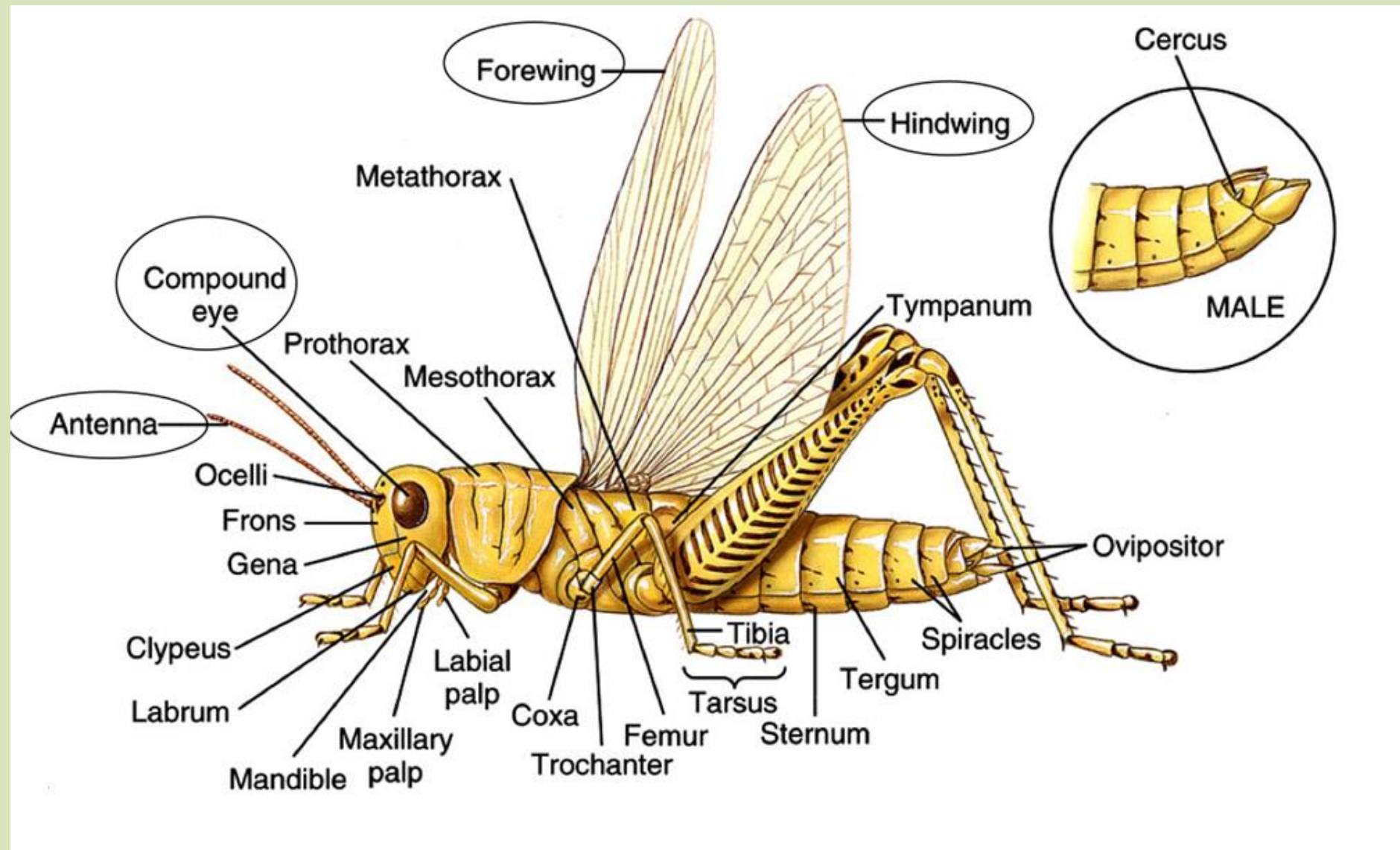


The general insect body plan

- They display tremendous diversity
- Thus we discuss using a hypothetical generalized insect

Tagmata

- Segments of the insect body are grouped into 3 well define regions – tagmata
 1. head
 2. thorax
 3. abdomen

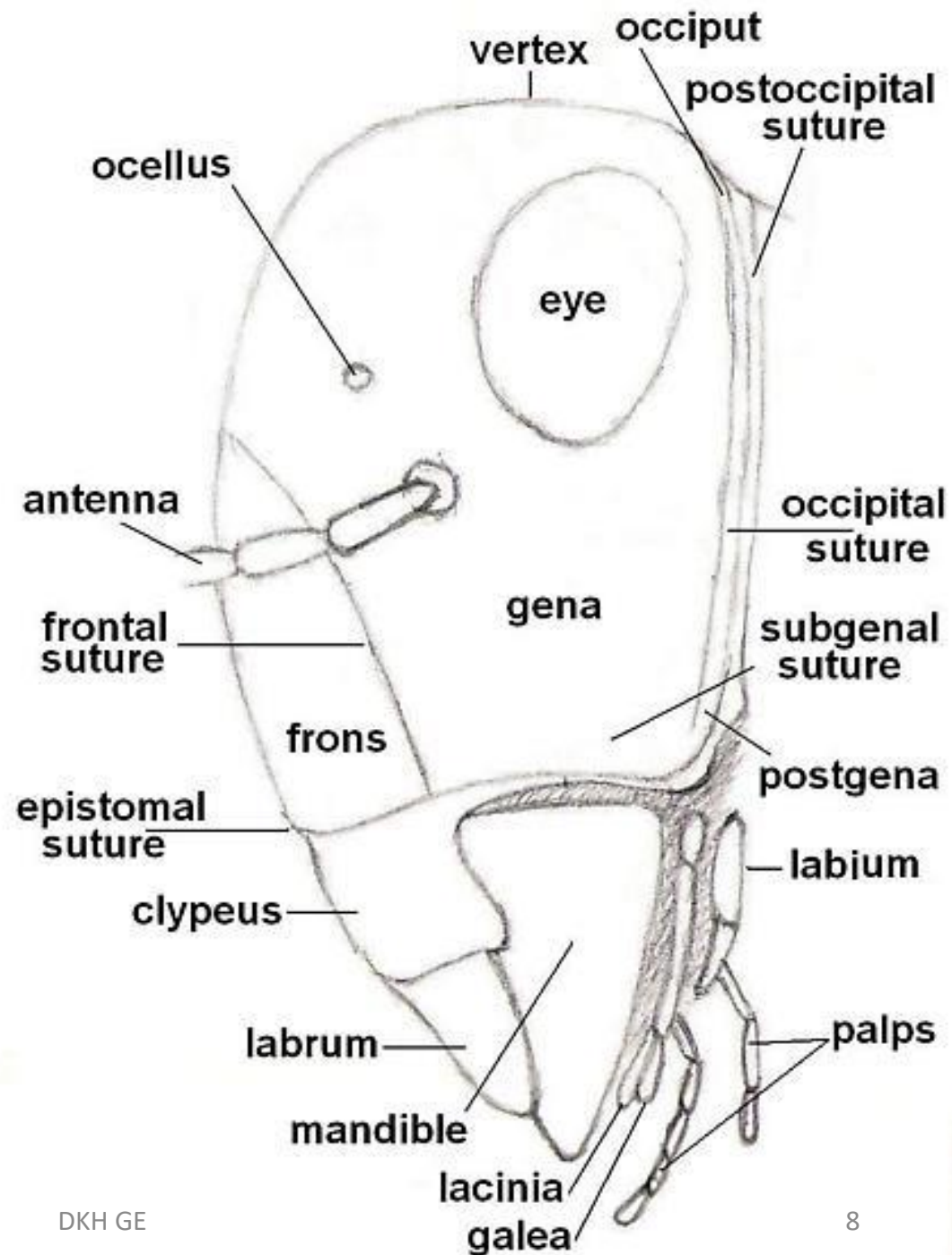


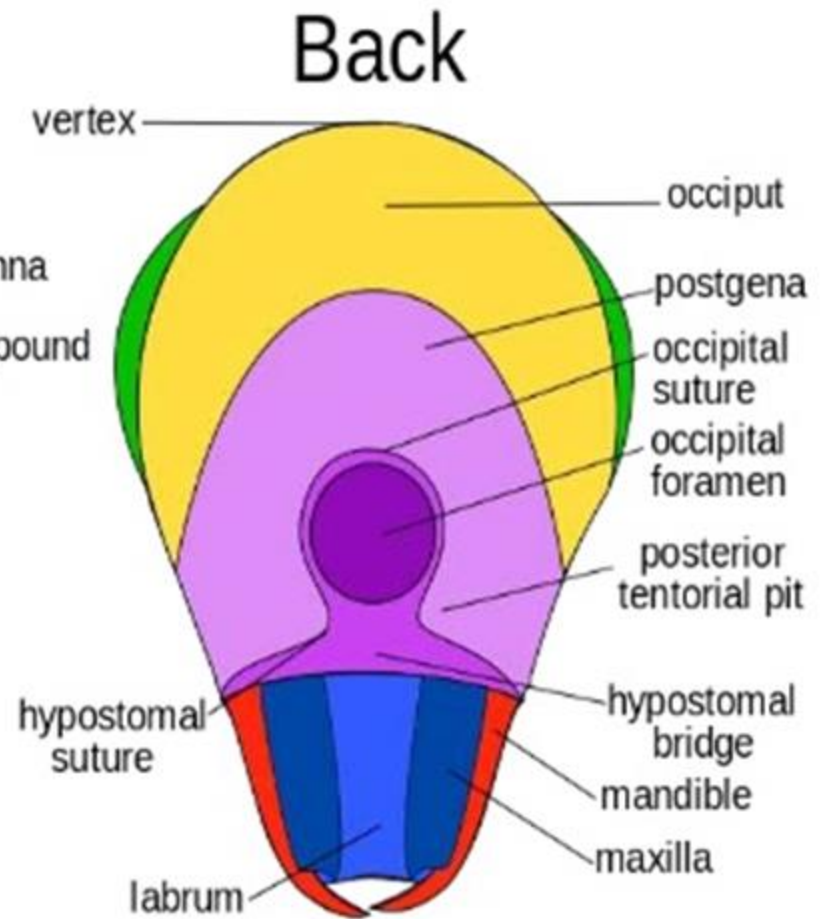
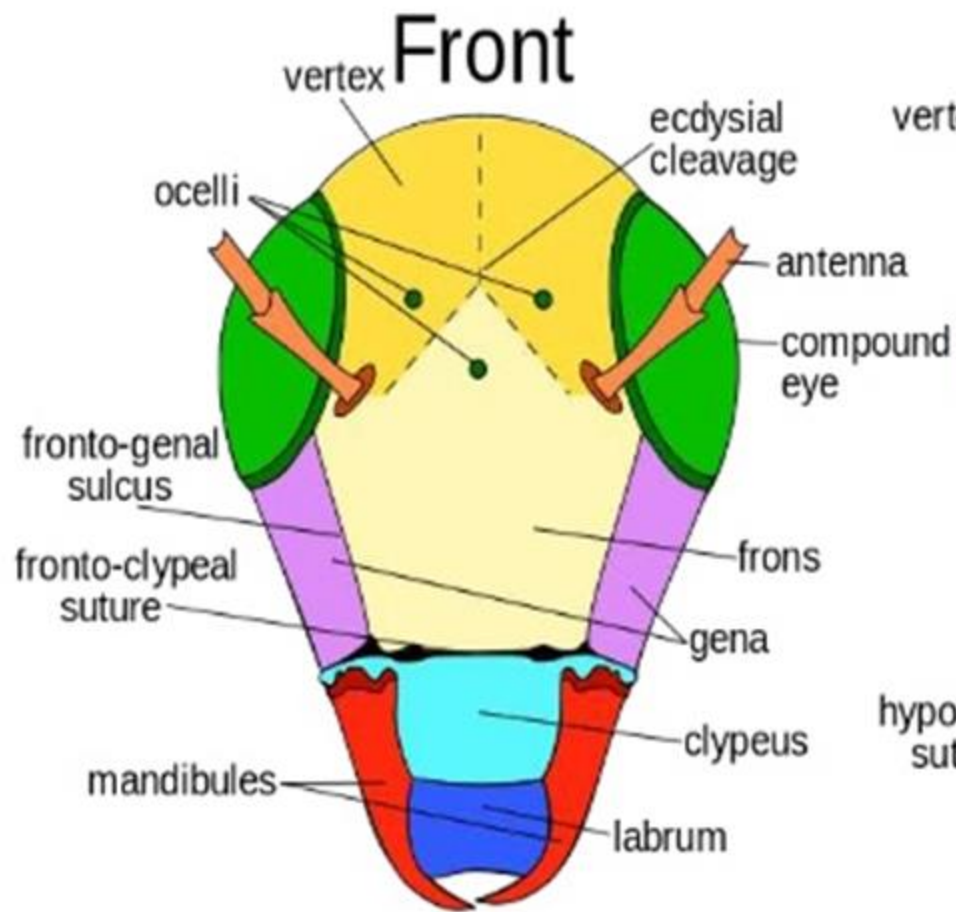
External features of an insect

Head

- The most anterior region or tagma of the insect body is the head.
- The head of Insect consist of six segments, although it looks like a single segment.
- This is because the six segments have fused so completely that the external evidence of their presence or origin is lost

- Frons
- Clypeus
- Labrum
- Gena
- Vertex
- occiput

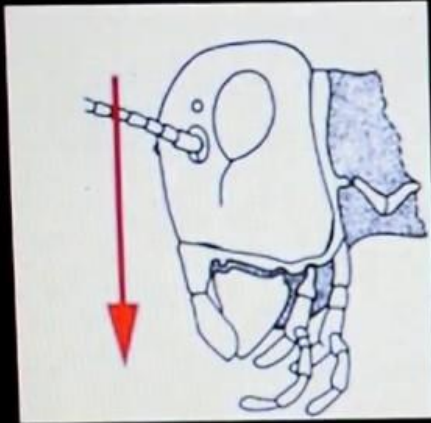




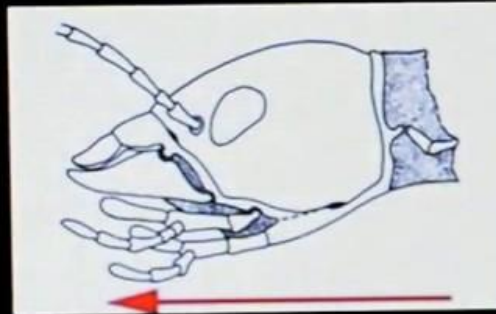
- Cranium is divided into various regions by sutures
- These structures are apparent in some insects while absent/ lacking in others
- The head is attached to the thorax by a membranous region – neck/cervix
- Cervical membrane is flexible
- Bears two pairs of lateral plates – the cervical sclerites

- Depending on the **inclination of the long axis** and the **position of the mouthparts** head is divided into 3 types;
 1. **Hypognathous** – long axis is vertical. Mouthparts ventral. e.g. grasshopper, cockroach etc.
 2. **Prognathous** – long axis horizontal. Mouthparts are anterior in position. e.g. Stick insect, soldier caste of termites etc.
 3. **Opisthognathous** – incline occipital foramen or it may retain transverse . Ventral region of the head is elongated. e.g. Red cotton bug.

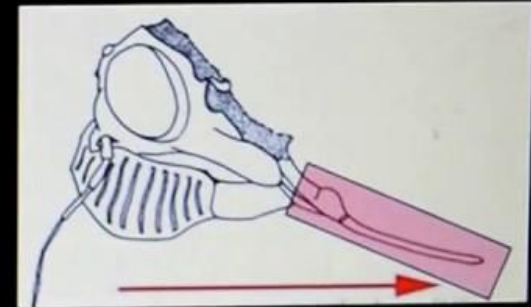
Hypognathous



Prognathous

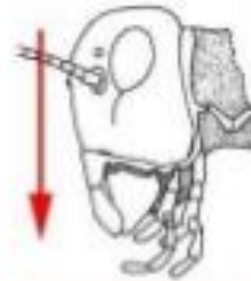


Opisthognathous

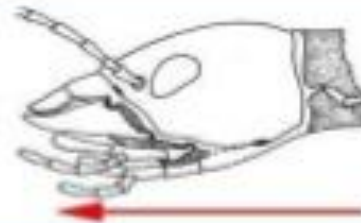


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Hypognathous: Mouthparts hang ventrally from the head capsule . This is the commonest orientation among plant feeders with chewing mouthparts .



Prognathous: Mouthparts positioned anteriorly to the head capsule. This is commonest among predatory insects.



Opisthognathous: Mouthparts positioned ventroposteriorly (rearwards) relative to the head capsule.

(a) hinged, so that they can be swung forwards; commonest among predators with piercing-sucking mouthparts, and (b) fixed, commonest among plant feeders with piercing-sucking mouthparts.



Compound Eyes

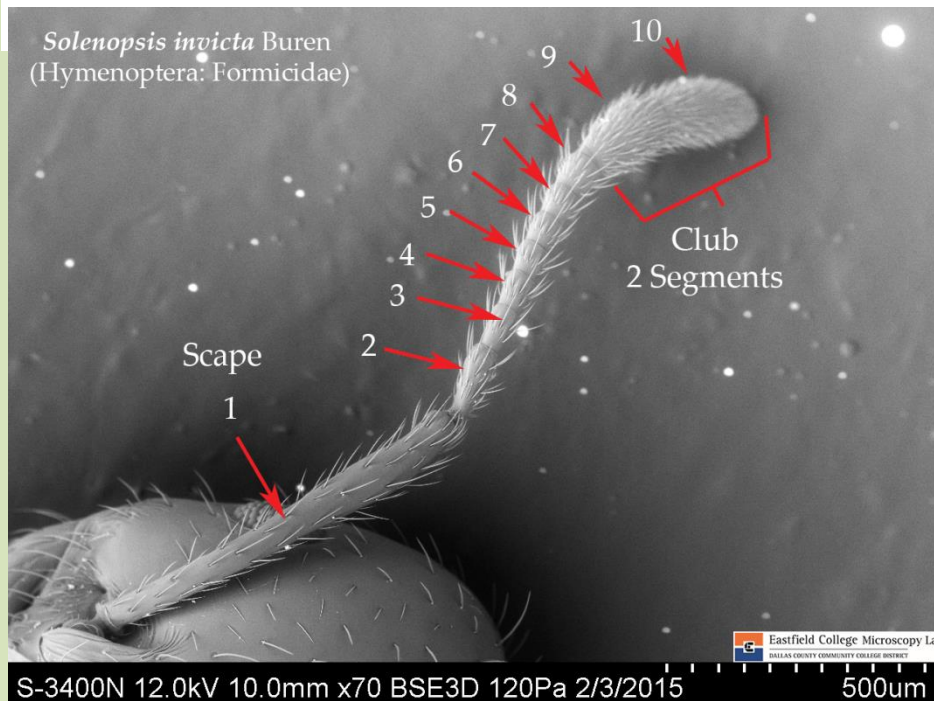
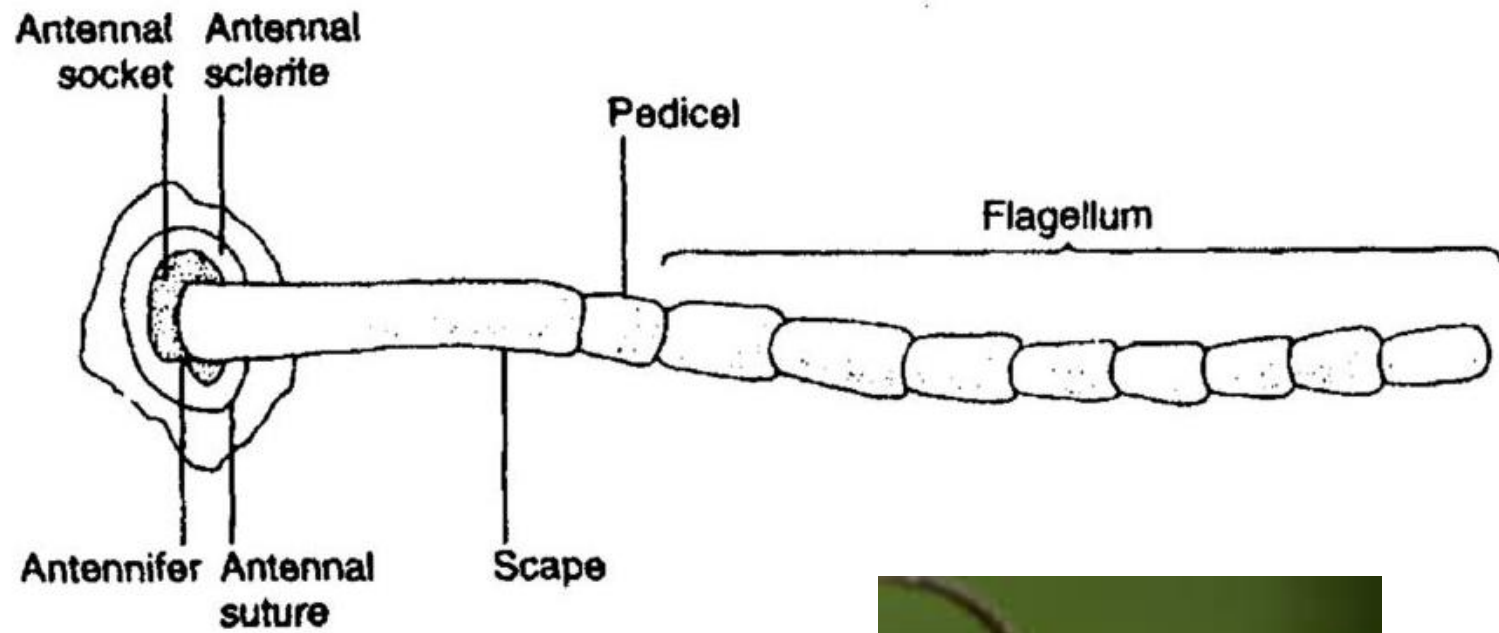
- on each side of the head
- Composed of many individual units – ommatidium
- Surface is divided into large number of hexagonal facets
- Facets serve as corneal lense
- Dorsal ocelli / simple eyes – three located on the anterior portion of the cranium

(TBC..)



Antennae

- All adult insects bear a pair of movable sensory appendages – antennae
- on the head between the compound eyes
- Segmented structure
- Three basic parts:
 1. scape – longer than other segments
 2. the pedicel – Johnson's organ is situated: insects can hear sounds
 3. multi-segmented flagellum
- Scape articulate the head with an antennal socket
- Have a sensory function – find way, detect danger, find mate, communicate, find food, olfaction, humidity detection, catching prey (mantid), holding opposite sex



- According to the shape and size (mostly in flagellum) there are different types of antennae

1. Filiform – threadlike

- All segments are about the same thickness
- Cockroaches (Order Blattodia), Grasshopper (Order Orthoptera) Longhorned Beetles (Order Coleoptera)

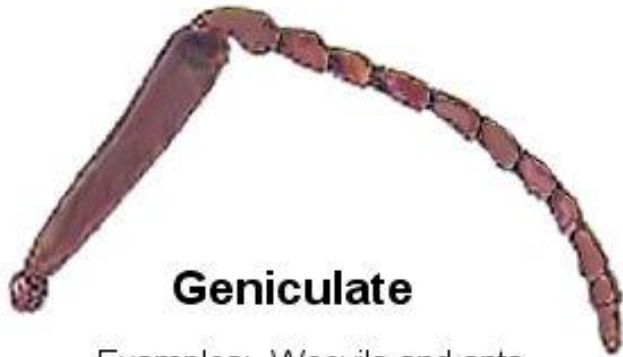


Filiform



2. Geniculate – elbowlike

- Hinged or bent like an elbow
- Bent abruptly at an angle at the distal end of scape forming a knee or elbow
- Bees and ants (Order Hymenoptera), Weevils (Order Coleoptera)



Examples: Weevils and ants



3. Clavate – club shaped

- Gradually clubbed at the end
- Carrion beetles (Order Coleoptera)



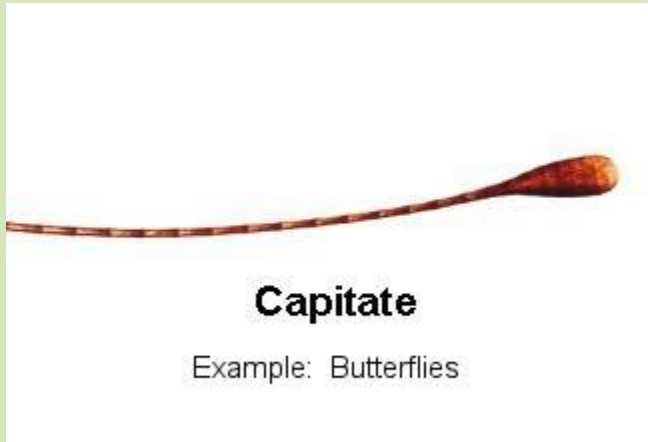
Clavate

Example: Carrion beetles



4. Capitate – knobbed or head shaped

- Abruptly clubbed at the end
- Antennae starts narrow at the base and right near the tips get bigger
- sap beetles (Order Coleoptera), butterfly (Order Lepidoptera)



5. Pectinate – comb shaped

- Segments are longer on one side
- Giving it a comb-like shape
- Fire-colored beetles and fireflies (Order Coleoptera)/



Pectinate

Examples: Sawflies and dobsonflies

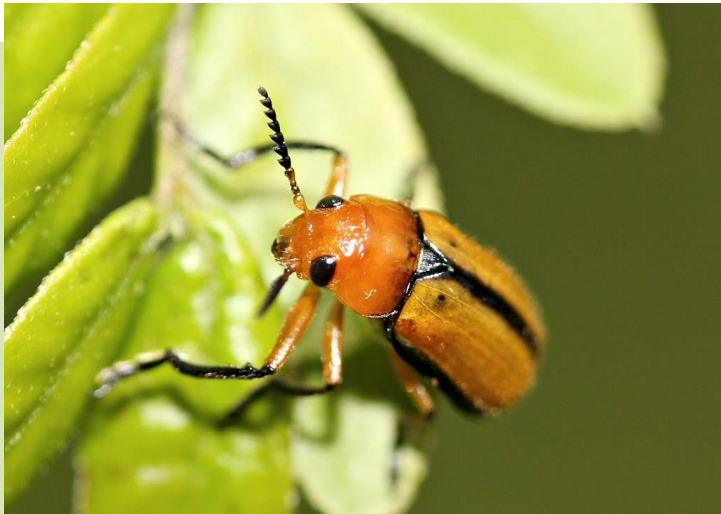


6. Serrate – saw shaped

- Saw-toothed shape
- Click beetles and mango stem borer (Order Coleoptera),



Serrate



7. Plumose – feather like

- Segments produce bunch of hairs from each joint
- Also called pinnate or bipectinate
- Moths (Order Lepidoptera), mosquitoes (Order Diptera)



Plumose

Example: Mosquitoes



8. Setaceous – bristle like

- Size of segments decrease from the base to apex
- Dragonflies and damselflies (Order Odonata), mayflies (Order Ephemeroptera), Cicad (Order Hemiptera)



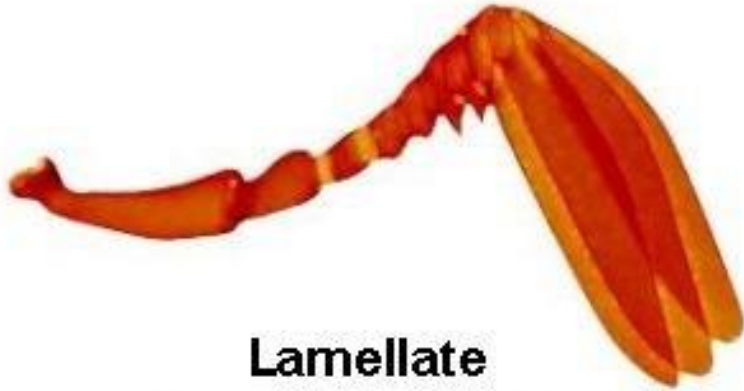
Setaceous

Examples: Dragonflies and cicadas



9. Lamellate – Plate like

- Last three segments extend in one side forming a leaf like structure
- Scarab beetles (Order Coleoptera)

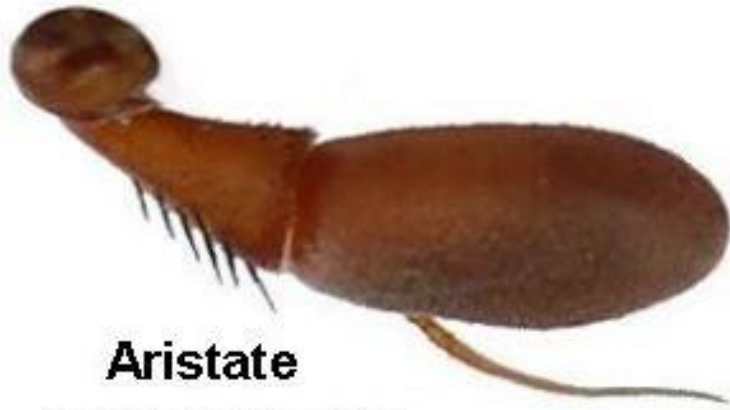


Lamellate
Example: Scarab beetles



10. Aristate – pouch like

- Pouch like with a lateral bristle
- House fly (Order Diptera)



Example: House flies



11. Monaliform – bead like

- Beadlike shape or a necklace like
- Termite (Order Isoptera)



Moniliform

Example: Termites



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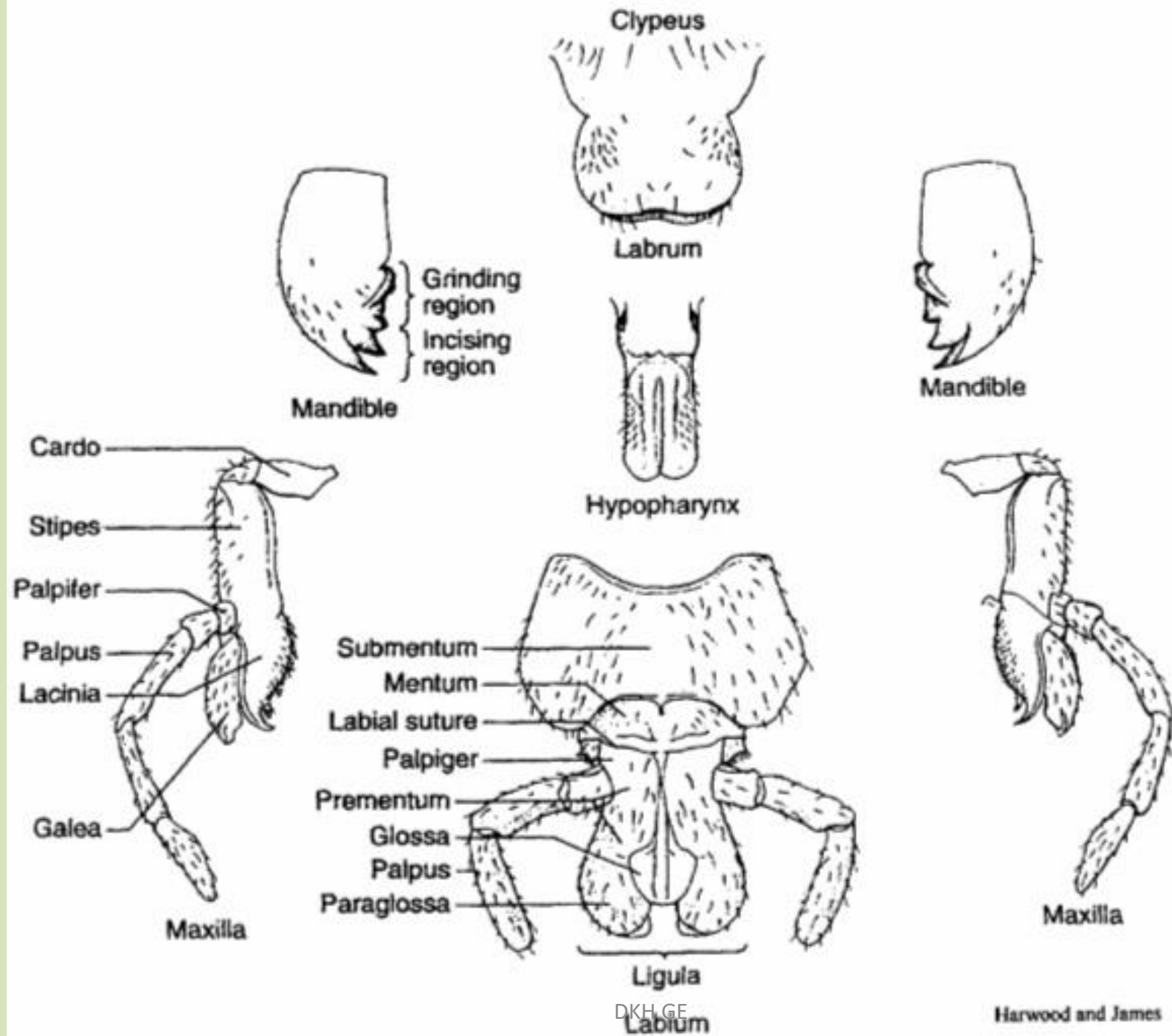
12. Bipectinate – feather like

- Have a feather shape
- moths (Order Lepidoptera)



Mouthparts

- Typical mouthparts consists of following;
 1. Labrum – anterior upper lip
 2. The hypopharynx
 3. A pair of mandibles
 4. A pair of maxillae
 5. Labium – posterior lower lip



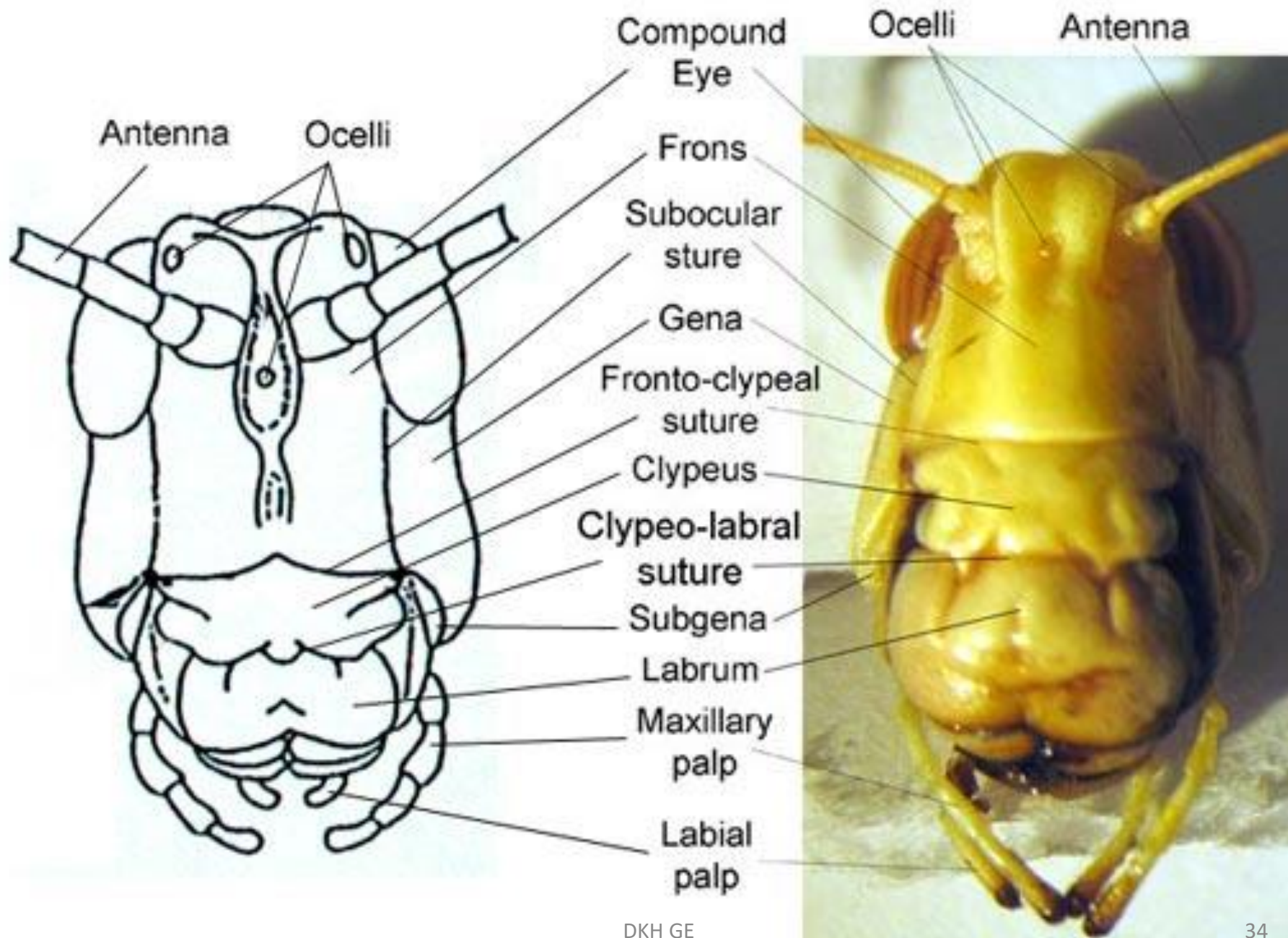
- If broadly categorizing there are two groups;
 - Chewing (mandibulates)
 - Sucking (haustellate)
- Five major types of mouthparts;
 - ✓ Chewing mouthparts- cockroach, grasshopper
 - ✓ Piercing and sucking- paddy bug
 - ✓ Siphoning mouthparts – butterfly
 - ✓ Sponging mouthparts- housefly
 - ✓ Chewing and lapping- honey bee

Chewing mouthparts

- Labrum – plate-like structure
- Located in front of other feeding structures protecting them
- Mandibles – pair of jaws
- Largest and most robust
- Used for masticating food
- Carnivores – serrated and knife-like, have piercing points
- Herbivores- broader and flatter

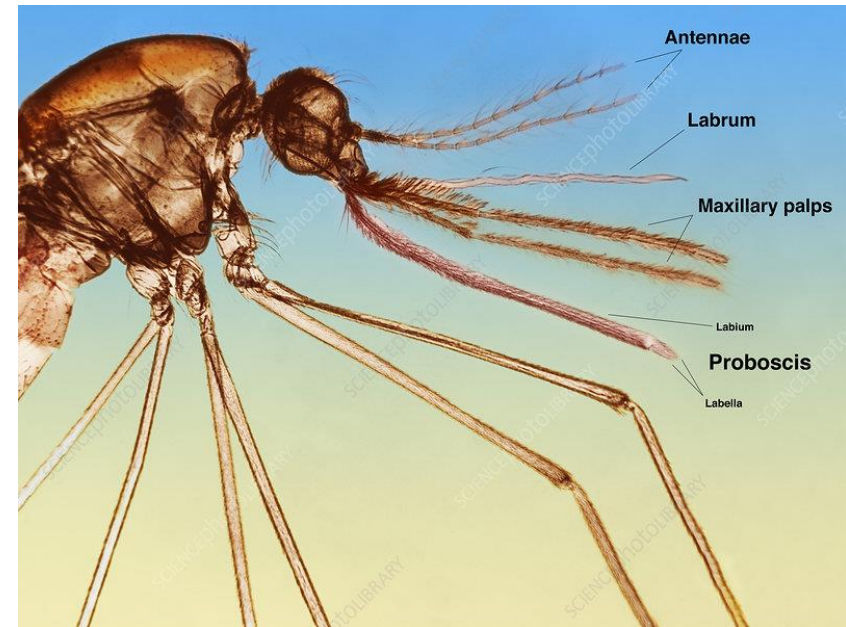
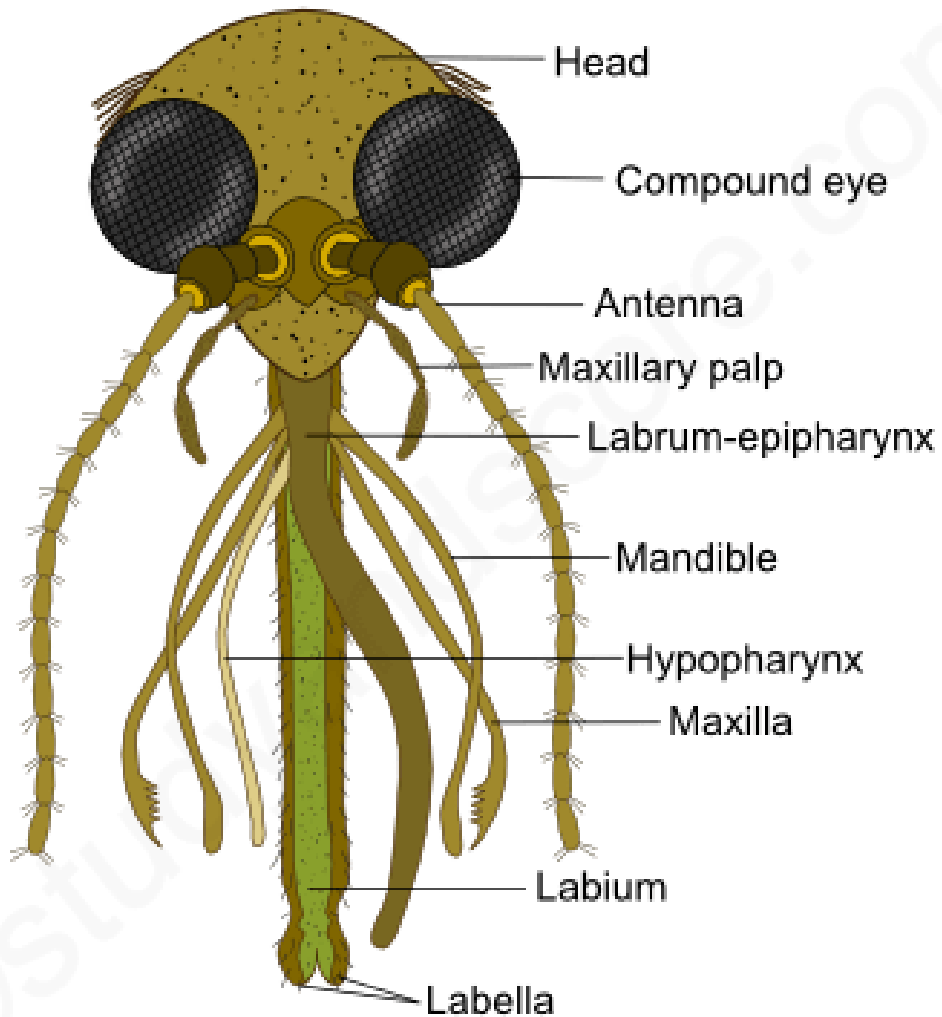
- Maxillae – act as fork and spoon to manipulate the food
- Bears lateral palps
- Serves as organs of touch and taste
- More mobile and less sclerotised
- Important in taking soft, liquid or food
- Hypopharynx – mix food with saliva
- Labium – fused two appendages





Piercing and sucking mouthparts

- Ex; aphids, mosquitoes, cicads, some bugs, etc.
- Mandibles & maxillae → modified into proboscis sheathed within a modified labium
- Mosquitoes (especially females) → all mouthparts are elongated
- Labrum forms the main feeding tube
- Paired mandibles and maxillae → form stylet – pierce animals



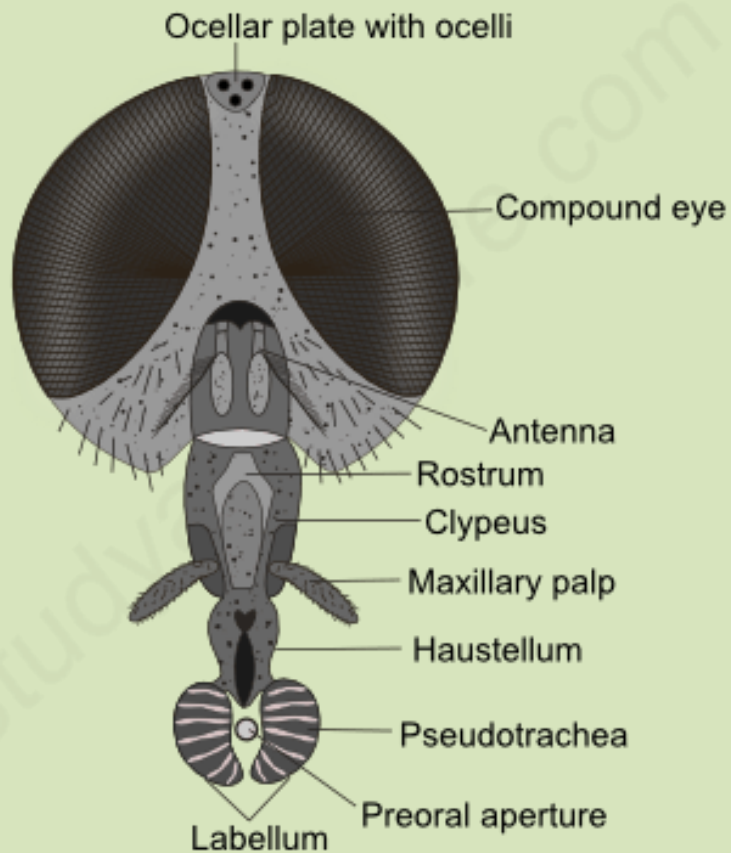
MOUTHPARTS OF MOSQUITO - PIERCING & CHEWING TYPE

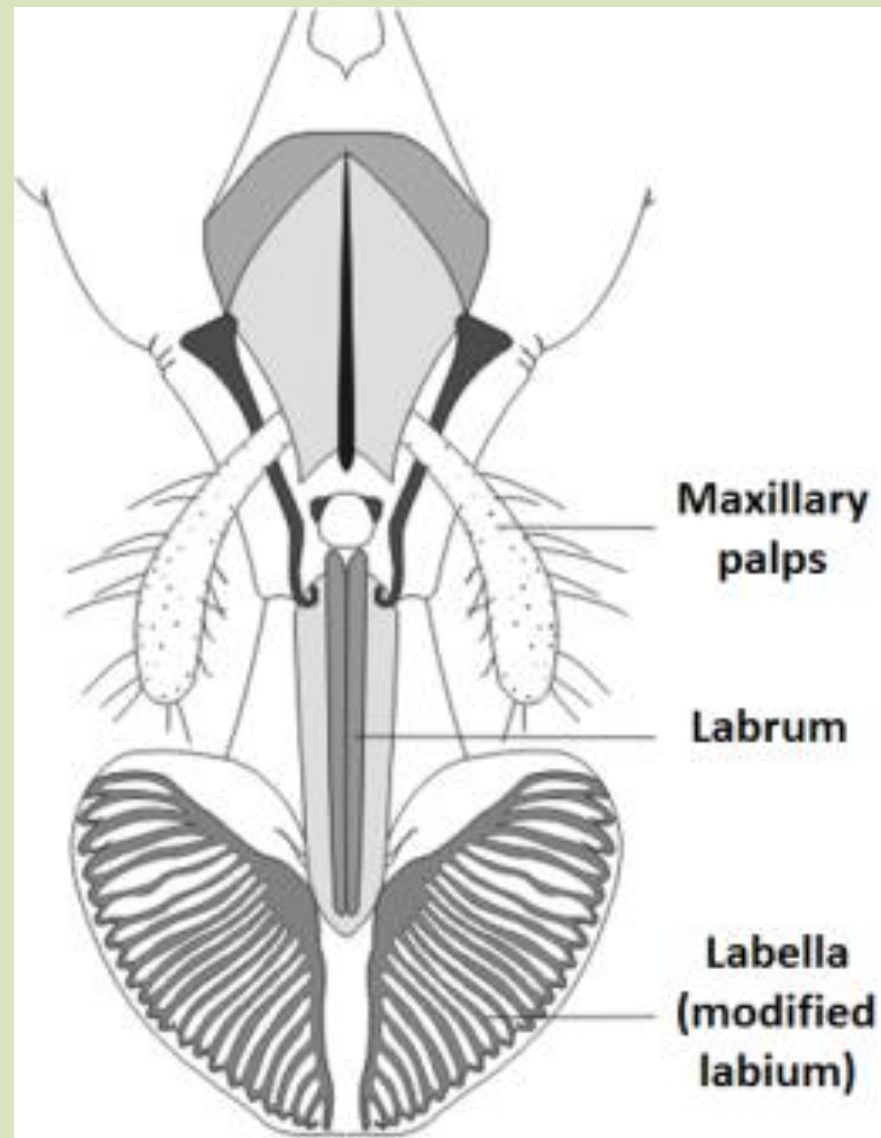
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Sponging mouthparts

- Ex; flies like house fly
- Sponge-like labellum modified from labium
- Paired mandibles and maxillae – much reduced and non-functional
- Maxillary palps are seen
- Labellums' surface → covered by minute food channels formed by the interlocking elongated hypopharynx and epipharynx → forms proboscis

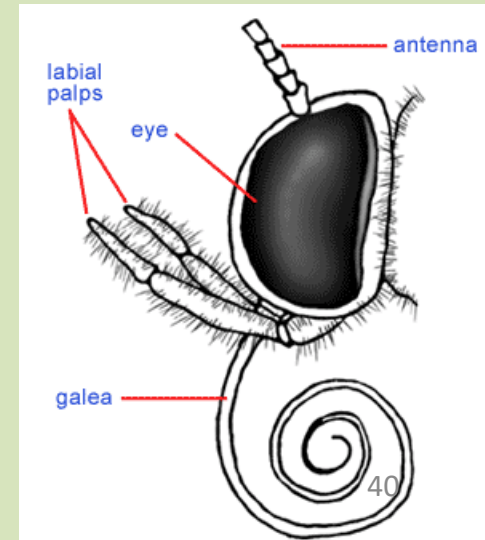
- Proboscis → channel liquid food to the oesophagus
- Food is drawn due to capillary action

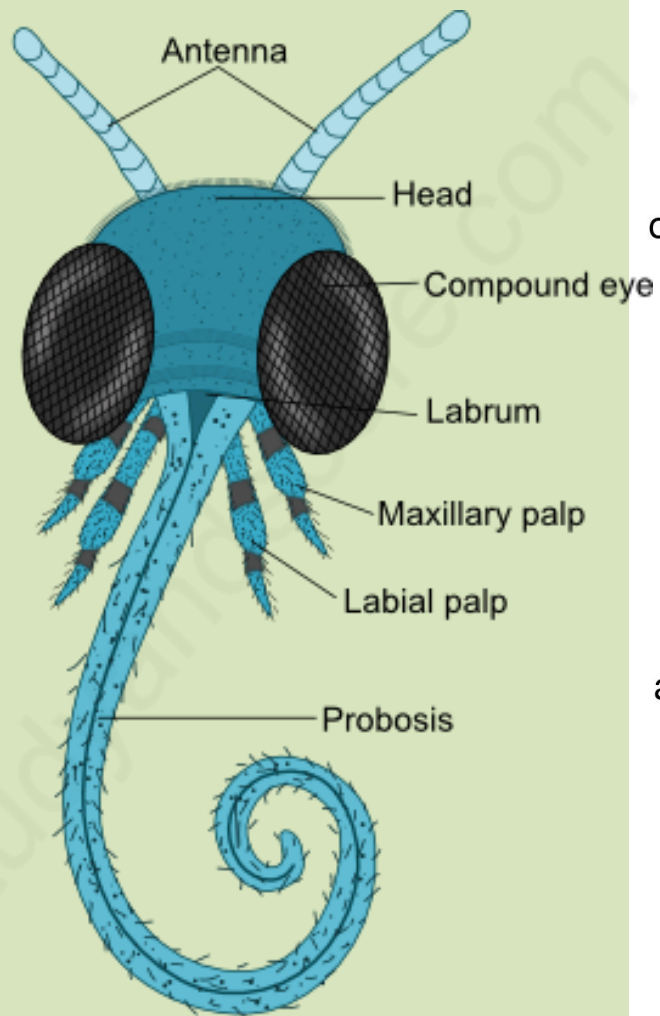




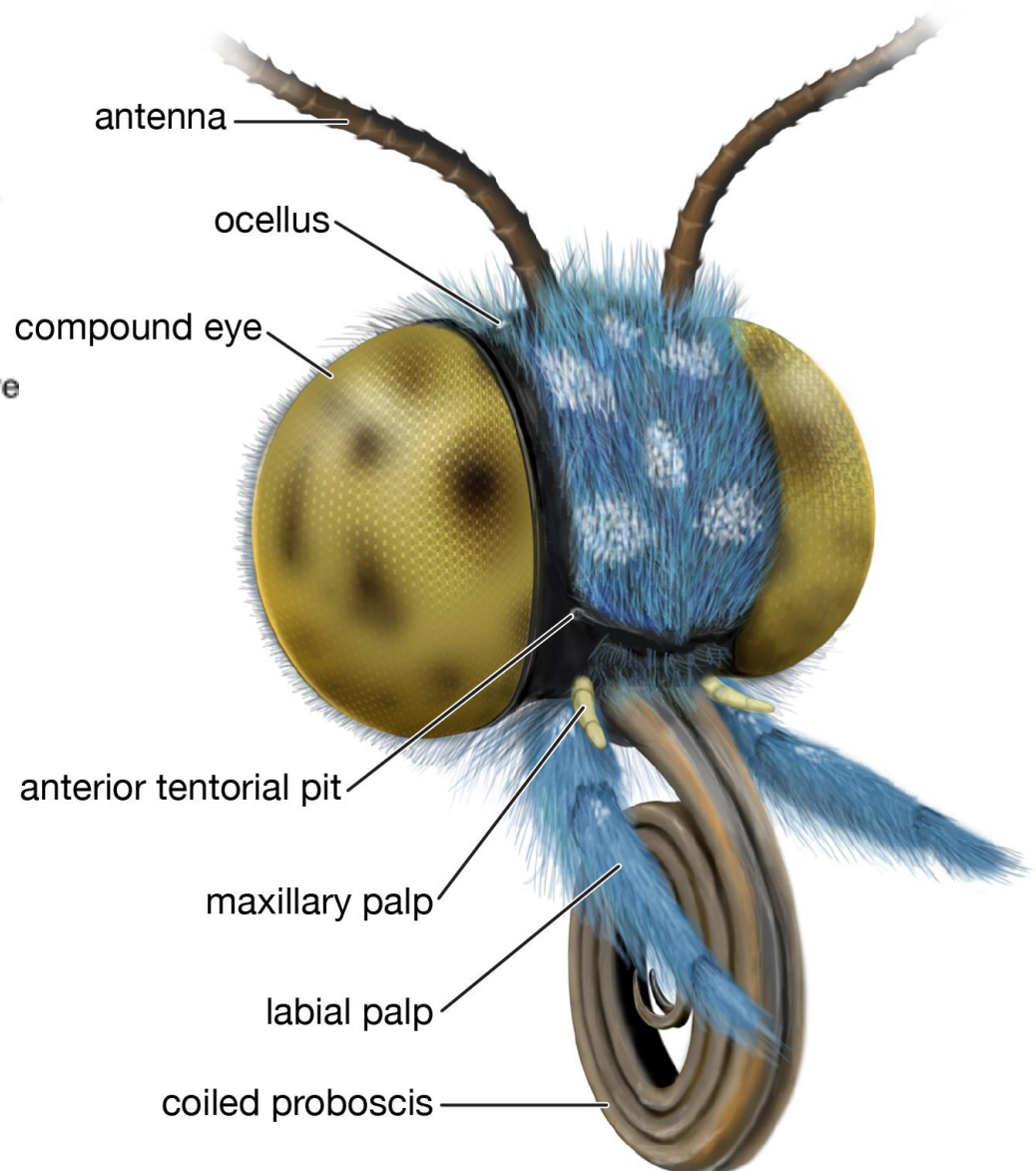
Siphoning mouthparts

- Ex; moths, butterflies
- Mandibles and labium → almost absent
- Only labial palps are visible
- Maxillary galeae developed forming a long proboscis.
- Proboscis can coil up





MOUTHPARTS OF BUTTERFLY - SIPHONING TYPE
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Chewing and lapping mouthparts

- Ex; bees
- Modified for collecting nectar and pollen
- consist of the labrum, epipharynx, mandibles, first pair of maxillae and labium
- Mandibles are short, smooth and spatulated situated one on either side of the labrum
- used in moulding wax and making the honeycomb

- The labium has reduced paraglossae
- The glossae are united and elongated to form the so called retractile tongue
- At its tip is a small labellum or honey spoon. The labial palps are elongated
- The glossae is used for gathering honey and it is an organ of touch and taste
- Maxillae bear small maxillary palps

- The galea and labial palps form a tube enclosing the glossae which moves up and down to collect nectar from flower nectaries.
- The nectar is sucked up through the tube, so formed, by the pumping action of the pharynx.
- The labrum and mandibles help in chewing the food

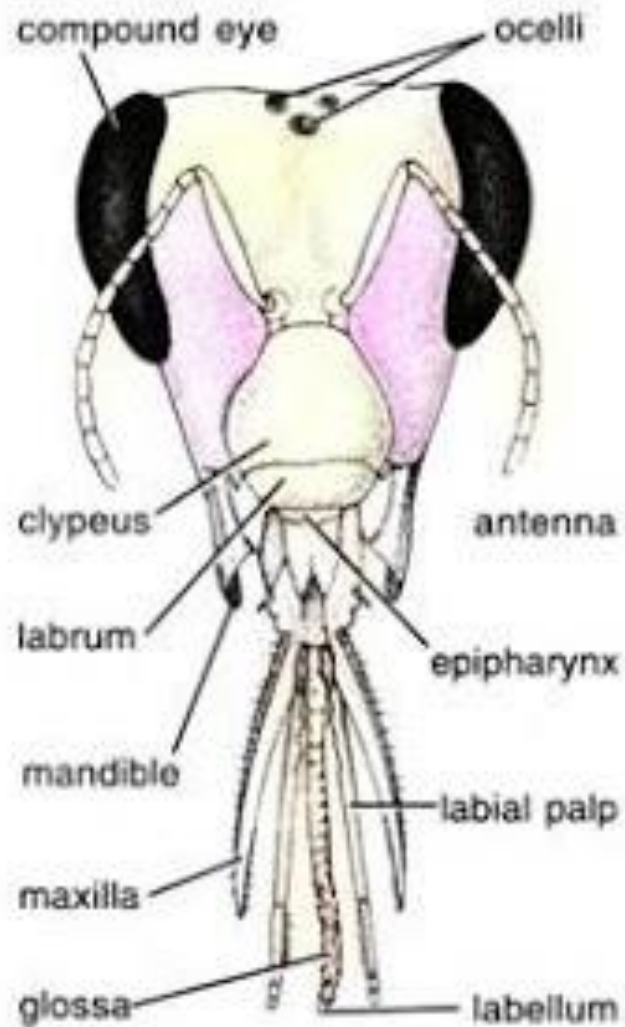


Fig. 77.3. Honeybee. Head and its appendages (Dorsal view)

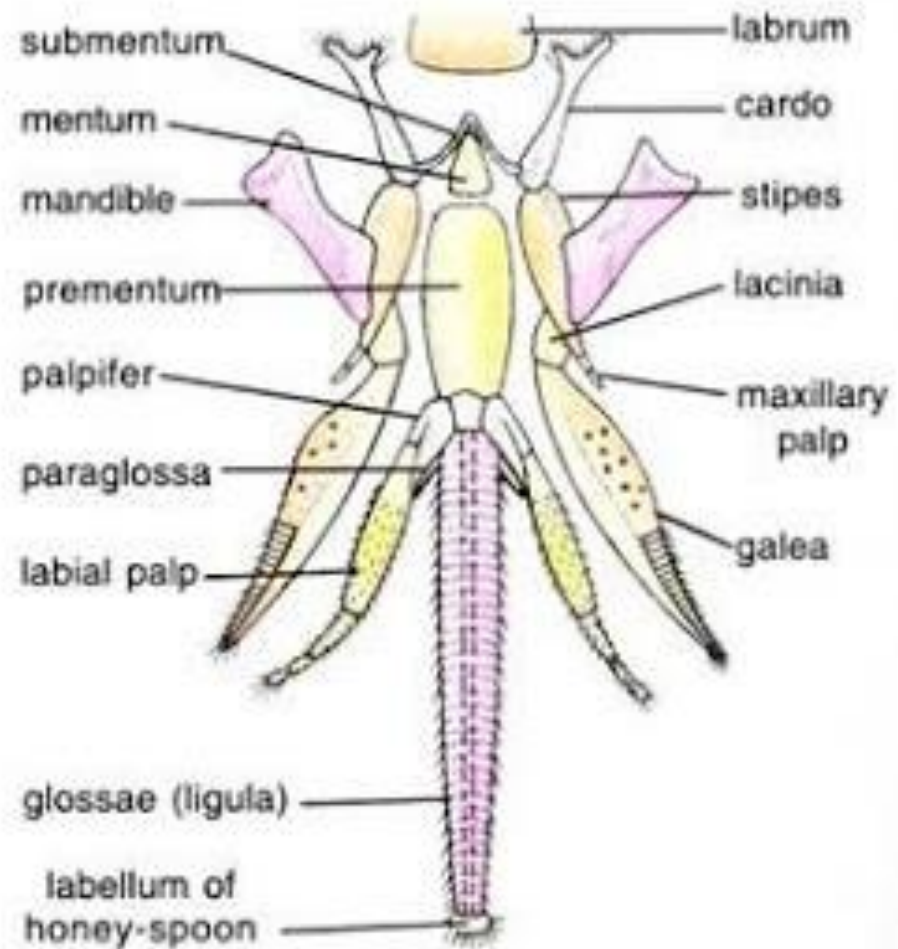


Fig. 77.4. Honeybee. Mouth parts.

THANK YOU!

